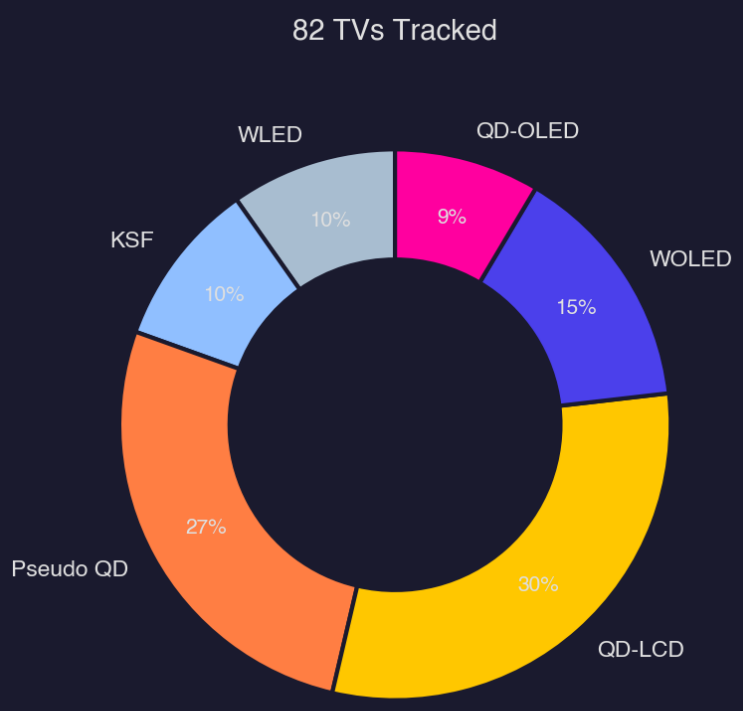
Display Technology Intelligence Report

February 2026

82 4K TVs tracked | 66 with pricing | 6 technologies | 3 8K excluded



1. Display Technology Overview

Section 1: Display Technology Overview

The display technology landscape remains stable this month, with 82 total TVs tracked across six major technologies and no new additions or removals from our database. This consistency provides a clear snapshot of the current competitive dynamics without the noise of monthly product launches disrupting performance comparisons.

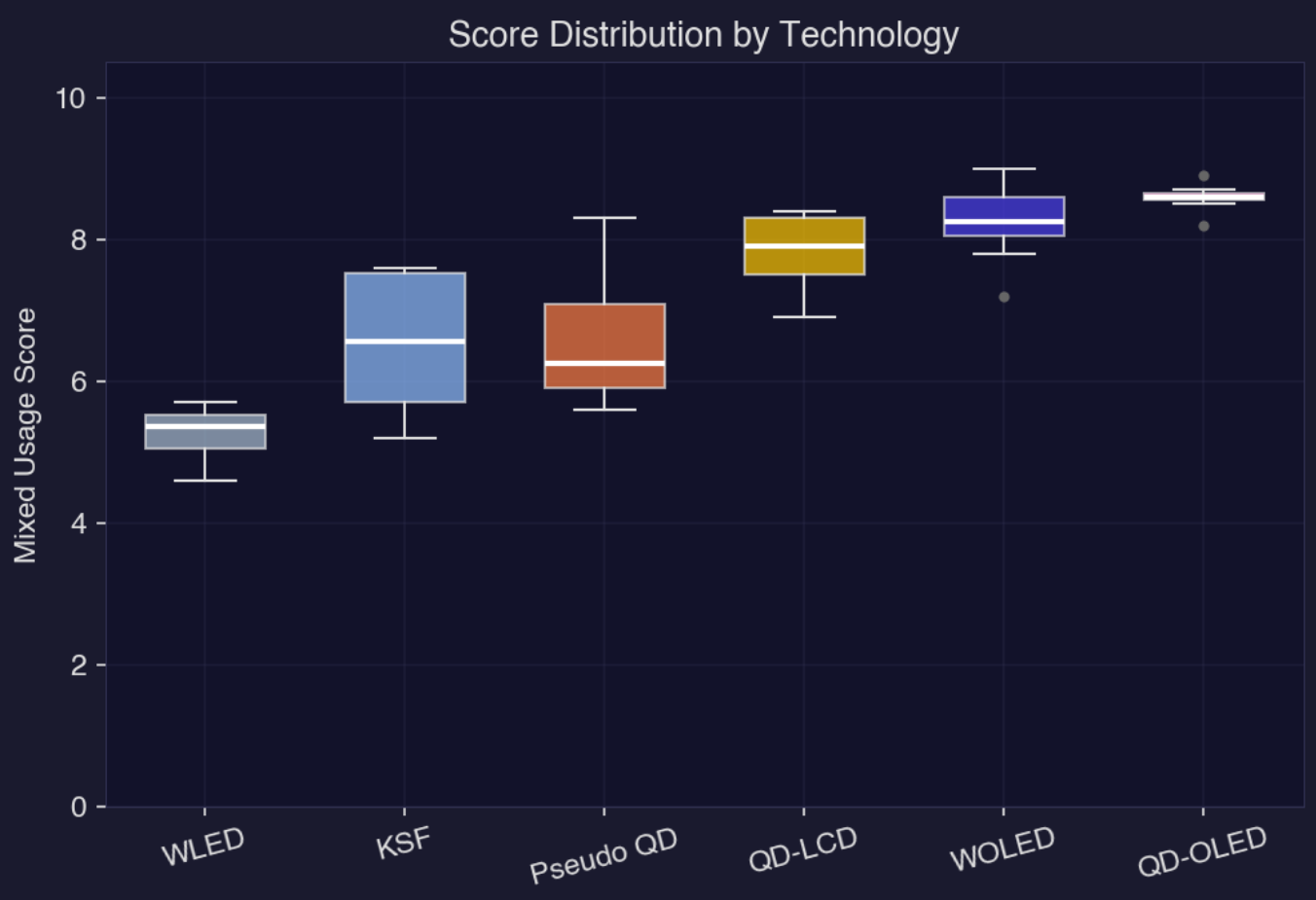
QD-OLED technology continues to dominate performance metrics, achieving the highest scores across all use cases with 8.6 for mixed usage, 8.8 for home theater, and an exceptional 8.9 for gaming. The technology maintains remarkable consistency, with its lowest-scoring model still hitting 8.2 for mixed usage. WOLED follows as the closest competitor, averaging 8.3 for mixed usage and 8.4 for home theater, though it trails QD-OLED by 0.5 points in gaming performance where QD-OLED's 8.9 average clearly leads the pack.

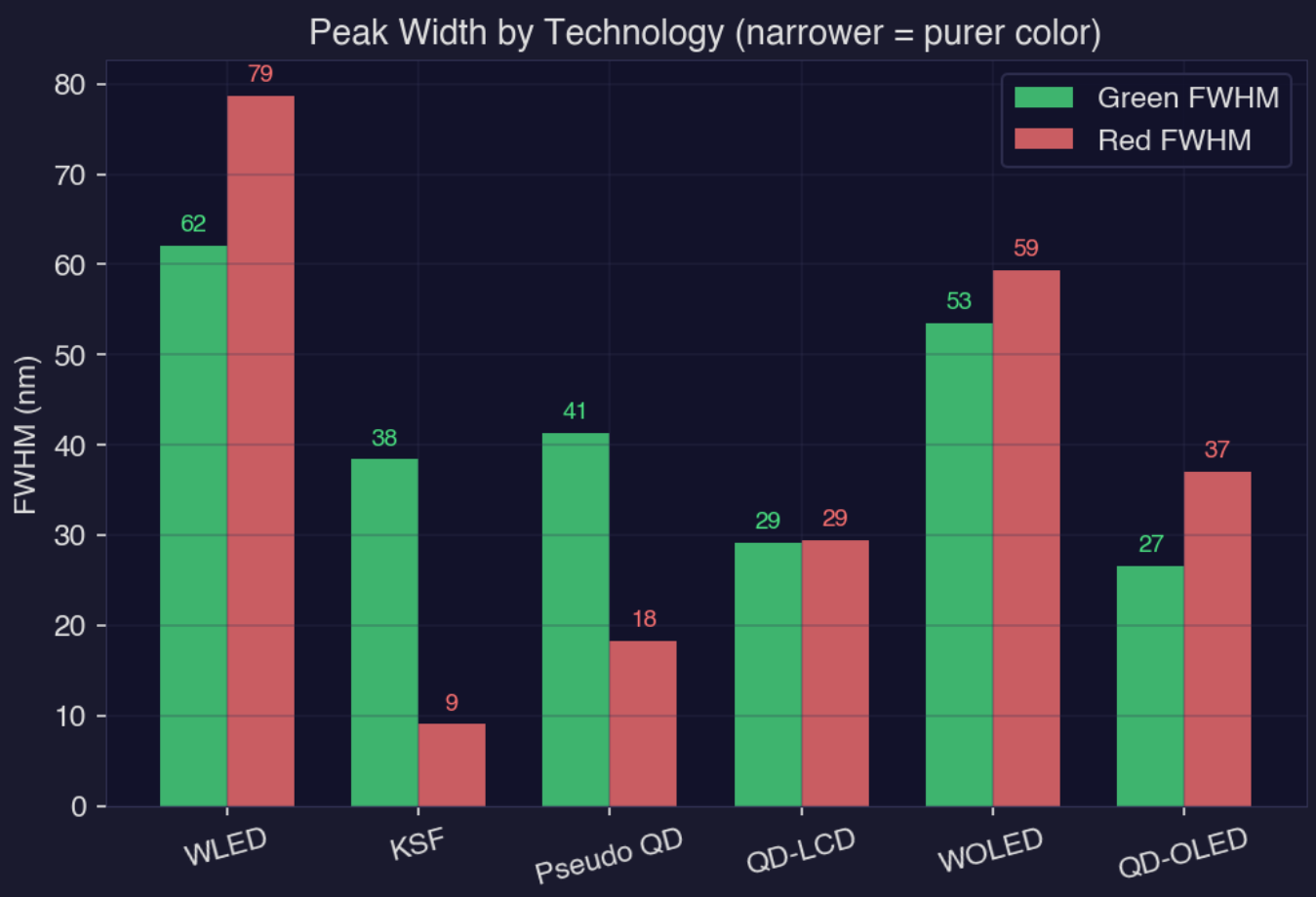
The performance gap between premium and mainstream technologies has crystallized into distinct tiers. QD-LCD occupies the performance middle ground with 7.8 mixed usage scores, significantly outpacing both Pseudo QD and KSF technologies, which cluster around 6.5-6.6. Traditional WLED technology anchors the bottom with 5.3 mixed usage scores, creating a 3.3-point spread between the best and worst performing technologies -- a gap that translates to meaningful real-world viewing differences.

Spectral analysis reveals the fundamental physics driving these performance differences. QD-OLED achieves superior color purity with 26.6nm green and 37.1nm red FWHM measurements, demonstrating the technology's ability to produce highly saturated colors. QD-LCD follows closely with 29.2nm green and 29.5nm red peaks, explaining its strong second-place showing among LCD technologies. The stark contrast becomes apparent with traditional WLED displays, where green peaks widen to 62.1nm and red peaks balloon to 78.7nm, creating the muddy color reproduction reflected in their lower performance scores.

KSF technology presents an interesting spectral profile with exceptional red purity at 9.1nm FWHM but broader green peaks at 38.4nm, suggesting manufacturers have optimized red phosphor efficiency while accepting compromises in green reproduction. WOLED shows relatively wide spectral peaks at 61.1nm green and 59.4nm red, yet still achieves strong performance scores, demonstrating how perfect black levels and infinite contrast can overcome color purity limitations.

The 40 models from 2025 currently dominate our dataset with 7.2 average scores across mixed usage and gaming categories, though this comparison lacks 2026 models for meaningful year-over-year analysis. This performance baseline will become crucial as 2026 models arrive to reveal whether manufacturers can push beyond current technology limitations or if we've reached a plateau in display innovation.


Score Distribution by Technology
Mixed Usage Score
10
8
6
4
2
0
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED


Peak Width by Technology (narrower = purer color)
FWHM (nm)
0
10
20
30
40
50
60
70
80
Green FWHM
Red FWHM
62
79
38
9
41
18
29
29
53
59
27
37
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED

2. Pricing Intelligence

Section 2: Pricing Intelligence

Current Pricing Landscape

The February 2026 display technology pricing hierarchy reveals a stark 5x spread between entry-level and premium technologies. WLED panels command the basement price of $380 median, while QD-OLED displays tower at $2,100 median -- establishing the clearest price-performance segmentation we've tracked. The middle tier splits cleanly between Pseudo QD at $700 and KSF phosphor conversion at $904, both targeting the volume sweet spot where consumers upgrade from basic edge-lit panels.

QD-LCD and WOLED technologies cluster in premium territory at $1,450 and $1,400 respectively, separated by just $50 despite fundamentally different manufacturing approaches. This pricing convergence suggests both Samsung's quantum dot LCD strategy and LG's white OLED approach have reached manufacturing maturity, with neither technology able to establish a meaningful cost advantage over the other.

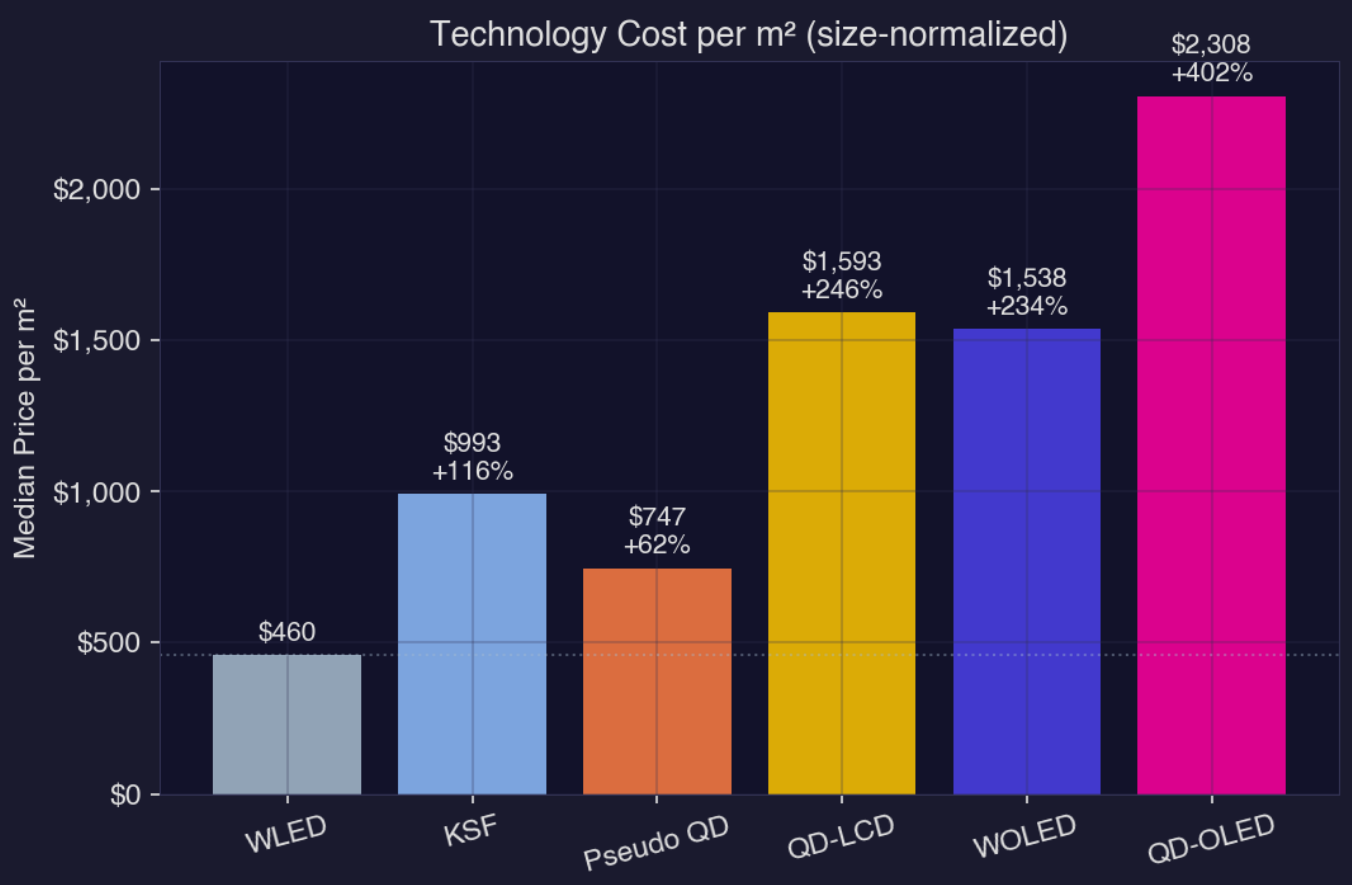
Per-square-meter analysis sharpens these distinctions further. QD-OLED extracts a brutal $2,308 per m2, commanding a 402% premium over baseline WLED pricing -- the steepest technology tax in television history. QD-LCD and WOLED demand nearly identical area premiums of 246% and 234% respectively, while KSF phosphor conversion doubles WLED pricing at a 116% premium.

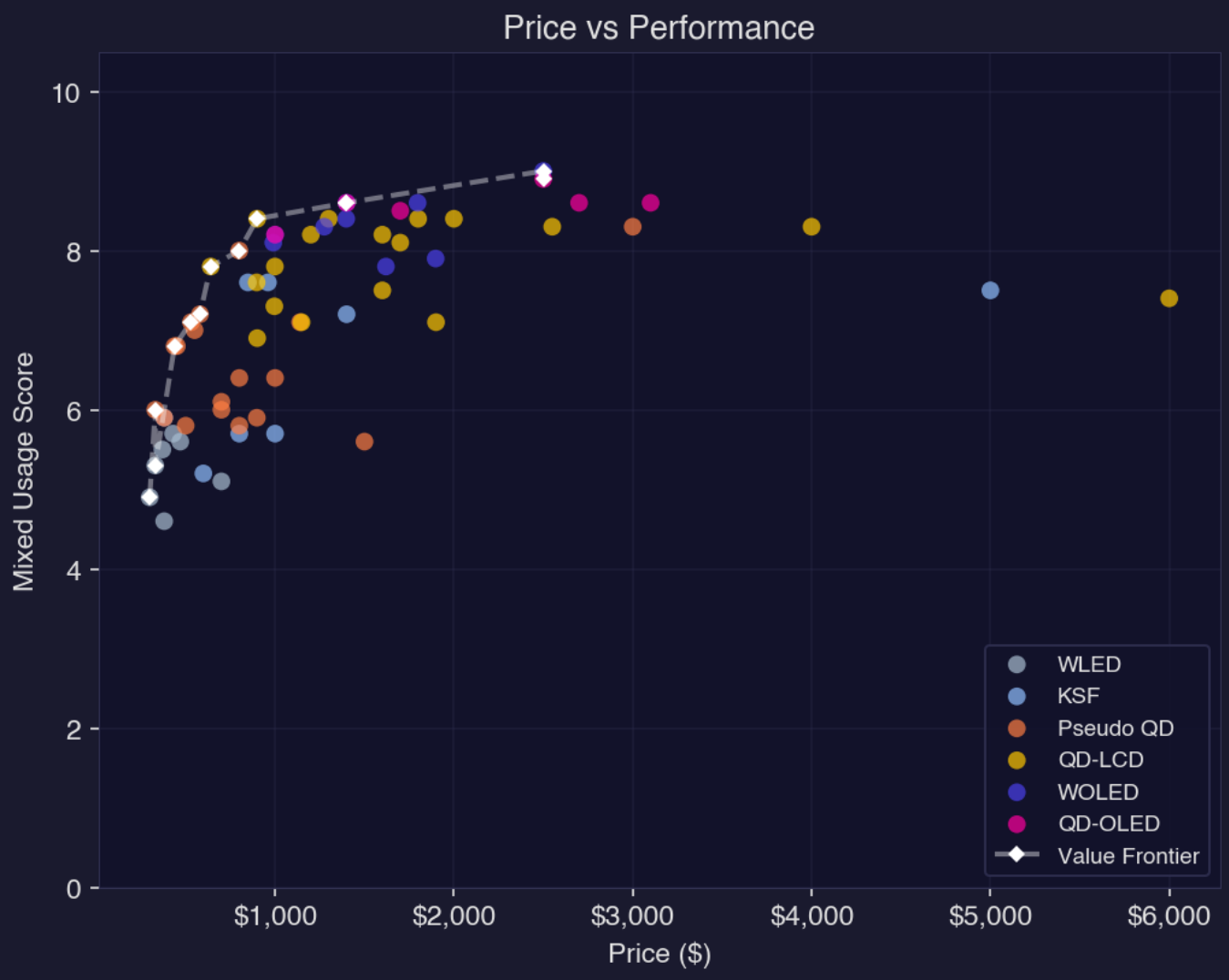
Value Performance Analysis

The value equation reveals Pseudo QD technology as the efficiency champion, with the Hisense QD6QF delivering 6.0 mixed-usage points for just $55 per point -- the lowest cost-per-performance ratio among premium technologies. This Hisense model undercuts traditional WLED value leaders by offering quantum dot color enhancement without the full manufacturing complexity of Samsung's QD-LCD approach.

WLED maintains its grip on absolute value territory through models like the LG UA77 at $61 per point, though its 4.9 mixed-usage score exposes the performance limitations that drive consumers toward quantum dot alternatives. The Samsung U7900F and TCL Q651G bracket the $62-64 per point range, with TCL's Pseudo QD implementation matching KSF phosphor performance from the S551G at identical $380 pricing.

The data exposes a critical value gap in premium territories -- no QD-LCD, WOLED, or QD-OLED models crack the top value rankings, suggesting these technologies sacrifice cost efficiency for absolute performance leadership. This February baseline establishes the pricing battleground where quantum dot variants will either justify their premiums through superior picture quality or face margin compression as manufacturing scales improve.


Technology Cost per m² (size-normalized)
Median Price per m²
$2,000
$1,500
$1,000
$500
$0
$460
$993
+116%
$747
+62%
$1,593
+246%
$1,538
+234%
$2,308
+402%
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED


Price vs Performance
Mixed Usage Score
10
8
6
4
2
0
$1,000
$2,000
$3,000
$4,000
$5,000
$6,000
Price ($)
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED
Value Frontier

3. Macro & Industry Context

Macro & Industry Context

The television display landscape has stratified into distinct technology tiers, with our 82-model database revealing how manufacturers deploy different solutions across price points. Samsung leads with 24 models spanning QD-LCD premiums and budget Pseudo QD offerings, while Hisense's 15 models demonstrate aggressive QD-LCD expansion in mid-tier segments. The technology distribution tells the story: QD-LCD commands 25 models as the new LCD flagship standard, Pseudo QD fills the mainstream with 22 models of rebranded KSF phosphor technology, and true OLED remains niche with 19 total models split between WOLED (12) and QD-OLED (7). TCL's 12 models and LG's 11 models round out brands that matter, while legacy WLED technology retreats to just 8 models in entry-level categories.

The technical differentiation between premium technologies has crystallized into measurable performance gaps. QD-OLED delivers the widest color gamuts and infinite contrast ratios but carries manufacturing complexity that limits Samsung and Sony to seven total models in our tracking. WOLED provides LG with cost-effective OLED production at scale, enabling broader model lineups despite slightly compressed color volumes compared to QD-OLED. QD-LCD represents the LCD evolution endpoint, offering 90%+ DCI-P3 coverage and local dimming zones that approach OLED-like contrast in premium implementations. Meanwhile, Pseudo QD marketing obscures what remains standard KSF phosphor technology with modest color improvements over basic WLED - a strategy that confuses consumers but maintains LCD margins.

Manufacturing realities drive the industry's next phase as tandem OLED structures promise brighter panels without efficiency penalties, potentially solving OLED's brightness limitations by 2027. MicroLED technology remains confined to ultra-premium displays above 100 inches, where pixel density requirements become manageable, but perovskite quantum dots could revolutionize color reproduction across all LCD categories within three years. Chinese manufacturers continue vertical integration strategies, with BOE and CSOT panel supply enabling Hisense and TCL to challenge Korean dominance through aggressive pricing on equivalent specifications. The QD-LCD category expansion signals LCD's final evolution before OLED manufacturing costs inevitably decline enough to eliminate the premium gap.

The display industry faces a watershed moment where QD-LCD represents LCD technology's peak performance just as OLED manufacturing scales toward mainstream pricing, setting up a 2026-2028 transition period that will determine which premium technology survives the cost compression cycle.